JAGANNATH UNIVERSITY

Department of Mathematics

4th Year 2nd Semester B.Sc. (Honours) (16th Batch)

Course Name: MATLAB Programming; Course code: MTHL 4211

Assignment

1. (a) Construct the following function for values of x from one to two in steps of 0.01 :

$$(i)\ y(x)=\frac{x^2}{x^3+1}\quad (ii)\ y(x)=\sin\left(\frac{x\cos x}{x^2+3x+1}\right)\quad (iii)\ y(x)=\frac{1}{x}+\frac{x^3}{x^4+5x\sin x}\quad (iv)\ y(x)=\frac{x}{x+1/x}.$$

(b) Evaluate the function $y=\frac{1}{x^3}+\frac{1}{x^2}+\frac{3}{x}$, for $x=-2$ to $x=-1$ in steps of 0.01.

(c) Write a MATLAB program that read two numbers x, y and display the value of x^y.

(d) Write a MATLAB program that read any number x and display the values of e^x and $\log x$.

2. (a) Write a MATLAB program, which calculate the sum of the series to five decimal places:

$$(i)\ \frac{1}{2}+\frac{2}{3}+\frac{3}{4}+\ldots\ldots\ldots+\frac{99}{100}\qquad (ii)\ \frac{1}{1^2}+\frac{3}{2^2}+\frac{5}{3^2}+\ldots\ldots\ldots+\frac{99}{50^2}$$

(b) Read a positive integer $N\geq 10$. Calculates to six decimal places of the sum and product:

$$1,\ 1+\frac{1}{2},\ 1+\frac{1}{2}+\frac{1}{3},\ldots\ldots\ldots,\ 1+\frac{1}{2}+\frac{1}{3}+\ldots\ldots+\frac{1}{N}.$$

(c) The constant π can be approximated by $\frac{\pi^2}{6}=1+\frac{1}{2^2}+\frac{1}{3^2}+\ldots\ldots$

Find the sum of the first 1000 terms. Hence find the value of π in six decimal places.

3. (a) Write a MATLAB program that takes a number from keyboard and tells whether it is positive or negative?

(b) Write a MATLAB program that take an integer number from keyboard and tells whether it is even or odd?

4. (a) Write a MATLAB program to find whether a number less than 1000 is prime or not.

(b) Write a MATLAB program which prints the first 25 prime numbers.

(c) Write a MATLAB program which prints the prime numbers between M and N and count the primes.

5. (a) The Fibonacci sequence is given by 1, 1, 2, 3, 5, 8, 13, …..

Write a MATLAB program which prints the first N terms of the Fibonacci sequence.

(b) Write a MATLAB program which prints the sum of the first 15 terms of the Fibonacci sequence.

6. (a) Input a positive number $N>9$. Write a MATLAB program to find whether N is prime or not. If the is not prime, find all of its divisors. Print the sum of the divisors. Hence verify the number is perfect or not.

(b) Input a positive number $5\leq N\leq 12$. Write a MATLAB program to find Factorial(N) or $N!$

(c) Stirling approximation is given by $n!\cong e^{-n}n^n\sqrt{2\pi n}$. Input two positive numbers $N, R\ (N\geq R)$. Write a MATLAB program to find the value of ${}^NC_R=\frac{N!}{R!(n-R)!}$.

7. Find the largest and smallest of $N>9$ integer values. Count the number of positive numbers; negative numbers; prime numbers and perfect numbers. Also calculate the sum of squares of the N values, average (mean), the variance and standard deviation of the N values.

8. (a) Write a MATLAB program to find the solution of the quadratic equation $ax^2+bx+c=0$. Hence obtain the solution of the equation $2x^2+6x+3=0$.

(b) Write a MATLAB program to find solution of the simultaneous equations $a_1x+b_1y=c_1,\ a_2x+b_2y=c_2$. Hence obtain the solution of the simultaneous equations $2x+5y=9,\ 5x-2y=8$.

MATLAB Programs for Numerical Methods:

1. Write a MATLAB program to find a real root of the equation $x^3 + x^2 - 1 = 0$ using Bisection method, Fixed point iteration, Newton-Raphson method and method of false position.
2. Interpolation (Newton's Forward, Newton's Backward and Lagrange's) methods.
3. Numerical integration (Trapezoidal, Simpson's 1/3, Simpson's 3/8 and Weddle's rule).
4. Numerical solution of differential equation (Taylor's method, Euler's method, Modified Euler's method, R-K method of order 4 and 4).
5. Gaussian Elimination method *with pivoting and without* pivoting to solve a system of linear equations.

Graphical Problem:

1. Use the comand **plot(*x*, *y*)** to plot the circle $x(t) = \sin t$, $y(t) = \cos t$, where t = linspace(0, 2*pi, 100). Also plot $y = t^2$.
2. Use the command **plot3(x, y, z)** to plot the circular helix $x(t) = \sin t,\ y(t) = \cos t,\ z(t) = t,\ 0 \le t \le 20$.
3. Plot $y = e^{-0.4x} \sin x$, taking 100 points in the interval $0 \le x \le 4\pi$.
4. Plot $y = e^{-0.4x} \sin x$, $0 \le x \le 4\pi$ taking 10, 50, 100 points in the interval.
5. Chapter – 6: Graphics (Page 161-190); Book: MATLAB 7: *RUDRAPRATAP*

Linear Algebra (Matrix related Problem):

1. Addition, subtraction and multiplication of matrices. Determinants, Elementary operations, determination of inverse of a Matrix, Eigen values, Eigenvectors, Characteristic polynomial, Cayley-Hamilton theorem verification, diagonalization. (See chapter-6; Book: *S.R. Otto and J.P. Denier*)
2. Solve Task problems (6.1-6.24), Book: *S.R. Otto and J.P. Denier*.
3. Compute $Y = Ax + B$, where x =linspace(0,10,4) , $A = \begin{pmatrix} 1 & 5 & 6 \\ 6 & 7 & 2 \\ 5 & 6 & 2 \\ 3 & 4 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 3 & 4 & 5 \\ 6 & 2 & 4 & 4 \\ 4 & 3 & 4 & 1 \end{pmatrix}$.
4. Compute $Z = A\sqrt{x}$, where x = linspace(0,10,5), $A = \begin{pmatrix} 1 & 5 & 6 \\ 2 & 9 & 3 \\ 1 & 2 & 0 \end{pmatrix}$ or $A = \sin x$.

Problems on Differential Equation:

1. Solve the following differential equation of motion of a nonlinear pendulum by MATLAB. Hence show the displacement and velocity vs. time plot of the pendulum in the same diagram with different graphic objectives. $\frac{d^2\theta}{dt^2} + \omega^2 \sin\theta = 0$, $\theta(0) = 1,\ \theta'(0) = 0.$ Use legend to identify each graphs.
2. Solve the differential equation

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 0$$

(See page 268; Book: *S.R. Otto and J.P. Denier*)

Subject to the initial conditions that $y(0) = 1$ and $y'(0) = 0$, from 0 to 1.

Miscellaneous Problems:

1. A vector is given by V = [5, 17, –3, 8, 0, –7, 12, 15, 20, –6, 6, 4, –7, 16]. Write a program as a script file that doubles the elements that are positive and are divisible by 3 or 5, and, raises to the power of 3 the elements that are negative but greater than –5.
2. A vector is given by x = [–3.5 5 –6.2 11.1 0 7 –9.5 2 15 –1 3 2.5]. Using conditional statements and loops, write a program that rearranges the elements of x in order from the smallest to the largest. Do not use MATLAB's built-in function sort.
3. The cost per mile for a rented vehicle is \$1.00 for the first 100 miles, \$0.80 for the next 200 miles, and \$0.70 for all miles in excess of 300 miles. Write MATLAB script files that determine the total cost and the average cost per mile for a given number of miles.

4. Body Mass Index (BMI) is a measure of obesity. In standard units it is calculated by the formula BMI=W/H^2, where W is weight in Kg, and H is height in meters. The obesity classification is:

BMI	Status
BMI <= 18.5	Underweight
18.5 < BMI <= 24.9	Normal
24.9 < BMI <= 29.9	Overweight
BMI >= 30	Obese

   Write a MATLAB program to calculate the body mass index with status of a person. The program asks the person to enter his or her weight (Kg) and height (m).
5. A vector is given by V = [5, 17, –3, 8, 0, –7, 12, 15, 20, –6, 6, 4, –7, 16]. Write a program as a script file that doubles the elements that are positive and are divisible by 3 or 5, and, raises to the power of 3 the elements that are negative but greater than –5.
6. A person in retirement is depositing \$300,000 in a saving account that pays 5% interest per year. The person plans to withdraw money from the account once a year. He starts by withdrawing \$25,000 after the first year, and in future years he increases the amount he withdraws according to the inflation rate. Calculate the number of years the money in the account will last assuming a constant yearly inflation rate of 2%. Make a plot that shows the yearly withdrawals and the balance of the account over the years. Use axis labels and legend also.
7. Write down the relation between Celsius and Fahrenheit temperature in degrees and construct a labeled table showing, vertically, the Fahrenheit equivalents of Celsius temperatures from 1 to 18 degrees in increments of two degrees.
8. Write a MATLAB program to calculate the interest on \$1000 at the rate of 6% at the end of each quarter in a year during five years period if it is computed quarterly.
9. The following were the daily maximum temperatures (in F) in Washington, DC, during the month of April 2002: 58 73 73 53 50 48 56 73 73 66 69 63 74 82 84 91 93 89 91 80 59 69 56 64 63 66 64 74 63 69. Use relational and logical operations to determine the following:
   a) The number of days the temperature was above 75.
   b) The number of days the temperature was between 65 and 80.
   c) The days of the month when the temperature was between 50 and 60.
10. Compute m in the the straight line $y = mx + c$, $4 \le x \le 22$, c =1.06.

Best of luck